



DEPARTMENT OF
COMPUTER SCIENCE



BOOLEAN EXPRESSIONS AND STATEMENTS

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LEARNING OBJECTIVES

At the end of the lesson, you should be able to:

- Illustrate the use of Boolean expressions in C.
- Illustrate the use of simple and compound statements in C.
- Discuss programme control in C.



SIMPLE STATEMENTS

We all have conversations with people daily, and while having these conversations, we make specific statements based on what the conversation is about.

Similarly, when we are trying to use the computer to solve problems, we must pass specific instructions or statements to the computer.



SIMPLE STATEMENT: EXAMPLE

For instance, if I want to use the computer to add three numbers, 5, 7, and 8, and print the result, using the C programming language, I can pass the instructions along with some lines of code.

```
int first_number = 5;
int second_number = 7;
int third_number = 8;
int result_of_addition = first_number +
    second_number + third_number;
printf("Result = %d \n", result_of_addition);
```



SIMPLE STATEMENT: EXAMPLE

```
int first_number = 5;
int second_number = 7;
int third_number = 8;
int result_of_addition = first_number +
    second_number + third_number;
printf("Result = %d \n", result_of_addition);
```

How many statements are in the code?

Can we have more than one statement on a single line?



COMPOUND STATEMENTS/BLOCKS

A simple statement contains only one expression and it ends with a semicolon.

A block contains more than one statement contained inside curly brackets. Let's see an example.

```
{  
    int first_number = 5;  
    int second_number = 7;  
    int third_number = 8;  
    int result_of_addition = first_number +  
        second_number + third_number;  
    printf("Result = %d \n", result_of_addition);  
}
```




COMPOUND STATEMENTS/BLOCKS

```
{  
    int first_number = 5;  
    int second_number = 7;  
    int third_number = 8;  
    int result_of_addition = first_number +  
        second_number + third_number;  
    printf("Result = %d \n", result_of_addition);  
}
```

The five statements are now grouped inside curly brackets.

The right curly bracket does not have a semicolon after it.



WHY COMPOUND STATEMENTS/BLOCKS

We need blocks because while trying to solve complex problems by breaking them into smaller ones, we need to treat solutions to the smaller problems as single units.



Why compound statements/blocks

The group of statements inside a block are treated like a single statement by the computer, which essentially means that solutions from different blocks can be seen as solutions to the smaller problems.



COMPOUND STATEMENTS/BLOCKS: APPLICATIONS

We use blocks when writing functions, decision making statements, and iterative statements.

Examples of decision making statements include the 'if', 'if-else' and 'else-if' statements. Also, examples of iterative statements include the 'for', 'while' and 'do-while' statements.



BOOLEAN AND BOOLEAN VARIABLES

Booleans represent values that are either true or false.

In C programming language, a Boolean variable is declared with the `bool` keyword and it can only take the values true or false.



BOOLEAN VARIABLES

It is noteworthy that Boolean values are returned as integers with;
1 (or any other number that is not 0) represents true
0 represents false.



COMPARING VALUES AND VARIABLES

To make decisions and find answers in programming, it is sometimes useful to compare values and variables. When this comparison activity takes place, the return value is usually a Boolean value.



PROGRAM CONTROL

In the C programming language, programme control is responsible for the sequential execution of a programme from one line to the next line until it reached the last line of the programme.



PROGRAM CONTROL

For some problem solving application, we want to alter this flow of operation to ensure that programme control moves to either a specific statement or block in our code.



FURTHER READING RESOURCES

- Bailey, T. (2005). An introduction to the c programming language and software design. July-2005
- Kernighan, B. W., & Ritchie, D. M. (2002). The C programming language.



SUMMARY

- A simple statement contains only one expression and it ends with a semicolon.
- A block contains more than one statement contained inside curly brackets
- Boolean values are returned as integers with; 1 (or any other number that is not 0) represents true 0 represents false.



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THANK
YOU